

$$R_1 \leftrightarrow R_2 \quad A = \begin{bmatrix} 2 & 4 & 0 \\ -1 & 3 & 2 \\ 1 & 2 & -3 \end{bmatrix} \quad \begin{array}{l} R_2 \leftarrow R_2 + \frac{1}{2}R_1 \\ = \\ R_3 \leftarrow R_3 - \frac{1}{2}R_1 \end{array} \quad \begin{bmatrix} 2 & 4 & 0 \\ 0 & 5 & 2 \\ 0 & 0 & -3 \end{bmatrix}$$

$$U^2 \begin{bmatrix} 2 & 4 & 0 \\ 0 & 5 & 2 \\ 0 & 0 & -3 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{cases} y_1 = 8 \\ -\frac{1}{2}y_1 + y_2 = -1 \\ \frac{1}{2}y_1 + y_3 = 7 \end{cases} \Rightarrow$$

$$\Rightarrow \begin{matrix} y_1 = 8 \\ y_2 = 3 \\ y_3 = 3 \end{matrix} \Rightarrow y = \begin{bmatrix} 8 \\ 3 \\ 3 \end{bmatrix} \quad U_x = y \Rightarrow \begin{bmatrix} 2 & 2 & 0 \\ 0 & 5 & 2 \\ 0 & 0 & -3 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$2) \begin{cases} 2x_1 + 4x_2 = 8 \\ 5x_2 + 2x_3 = 3 \\ -3x_3 = 3 \end{cases}$$

$$Z = \begin{cases} X_3 = -1 \\ X_2 = 1 \\ X_1 = 2 \end{cases}$$

c) (1.5 points) Find the remainder $R(f)$ of the quadrature formula in b), using Peano's theorem.

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pag 117
EXEMPLU

$$z) \begin{cases} ax_0 + b = 1 \\ a \geq 0 \end{cases} \Rightarrow \begin{cases} b = 1 \\ a = 0 \end{cases}$$

$$b_{00}(x) = 1$$

$$\Rightarrow \begin{cases} b_{1,1}(x_0) \geq 0 \\ b'_{1,1}(x_1) \geq 1 \end{cases} \Rightarrow \begin{cases} \hat{a}x_0 + b = 0 \\ a = 1 \end{cases}$$

$$2) \begin{cases} a = 1 \\ b = 1 \end{cases}$$

$$2) \quad h(x) = x -$$

$$\begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$$

$$B_2 f(x) = 1 \cdot f(-1) + (x+1)f'(1) = f'(1)x + f(-1) + f'(1)$$

$$f'(1)x + f(-1) + f'(1)$$

3) Let $f \in C^2[-1,1]$.

a) (1.5 points) Find $(Pf)(x)$, the polynomial interpolating the function f , given $f(-1)$ and $f'(1)$.

b) (1 point) Integrate $\int_{-1}^1 (Pf)(x) dx$ to obtain a quadrature formula for $\int_{-1}^1 f(x) dx$. Find its degree of precision.

c) (1.5 points) Find the remainder $R(f)$ of the quadrature formula in b), using Peano's theorem.

